

Continuous Assessment 2

Important Note:

1. Please submit PDF file instead of file in Word format.
2. This homework is to be completed within a stipulated deadline.
3. Please do online submission of the PDF file before or by **14 April 2026**.

Student Name: Stephen Manickam

Q1:

A robot is observing a wall of red color. The robot records the perceived values of the wall's color every one hour. One morning's outputs from the robot's eyes are as follows: (245, 2, 10), (254, 5, 9), (247, 6, 2), (243, 9, 1). What is the vector which describes the concept-physical representation of the wall's color?

mean or each pixel?

Your answer:

Your Solution: (write the details of your solution here)

$$\begin{bmatrix} 245 \\ 2 \\ 10 \end{bmatrix} \rightarrow 1^{\text{st}} \text{ hr} \quad \begin{bmatrix} 254 \\ 5 \\ 9 \end{bmatrix} \rightarrow 2^{\text{nd}} \text{ hr}$$

$$\begin{bmatrix} 247 \\ 6 \\ 2 \end{bmatrix} \rightarrow 3^{\text{rd}} \text{ hr} \quad \begin{bmatrix} 243 \\ 9 \\ 1 \end{bmatrix} \rightarrow 4^{\text{th}} \text{ hr}$$

$$\text{mean} \rightarrow \begin{bmatrix} 247.25 \\ 5.5 \\ 5.5 \end{bmatrix}$$

Q2:

There are multiple curves on a wall. The wall itself is a planar surface. On the wall, these curves can be categorized into three types: straight lines, circles, and cubic curves which are described by polynomial equations of degree 3. Design a neural network system which enables a robot to cognize and recognize these three types of curves.

Your answer:

Your Solution: (write the details of your solution here)

1) Sample Image using edge detection

2) represent Curve by a vector

Line: $y = ax + b$

Circle: $(x - x_0)^2 + (y - y_0)^2 = r^2$

Cubic: (polynomial of 3)

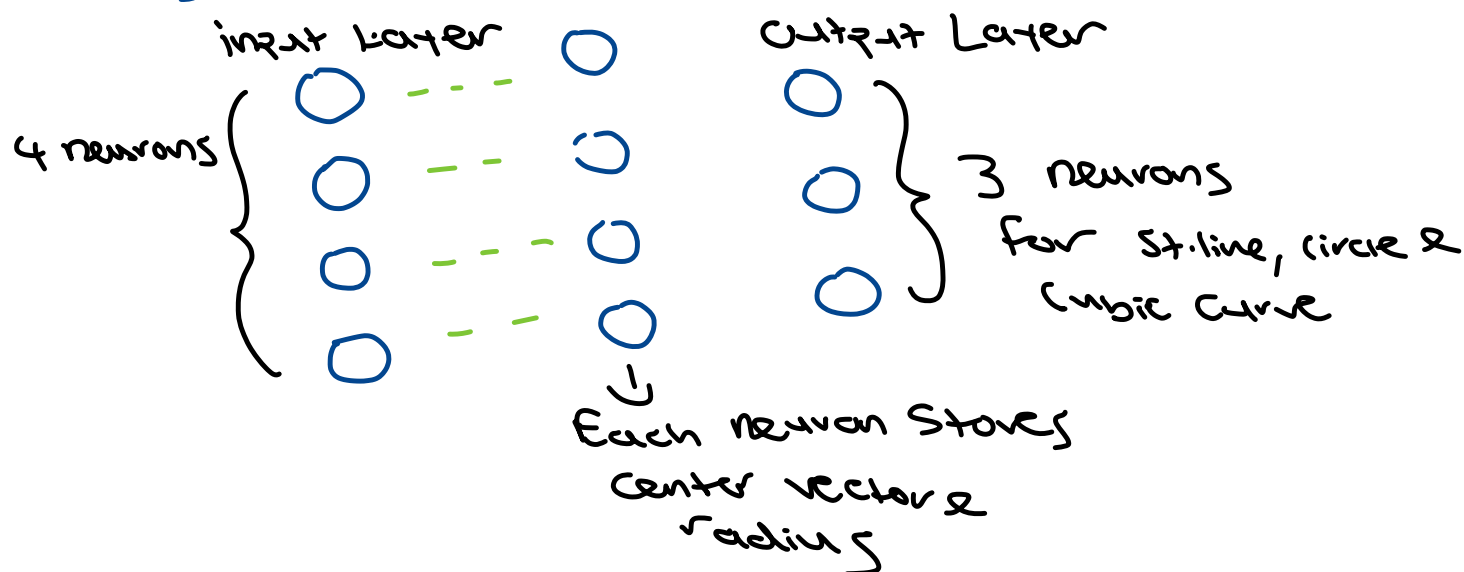
$y = ax^3 + bx^2 + cx + d$

Since Curves can be defined by equations, we will use this equation. So now, we only need 4 parameters.

→ Extract the parameters & form vector

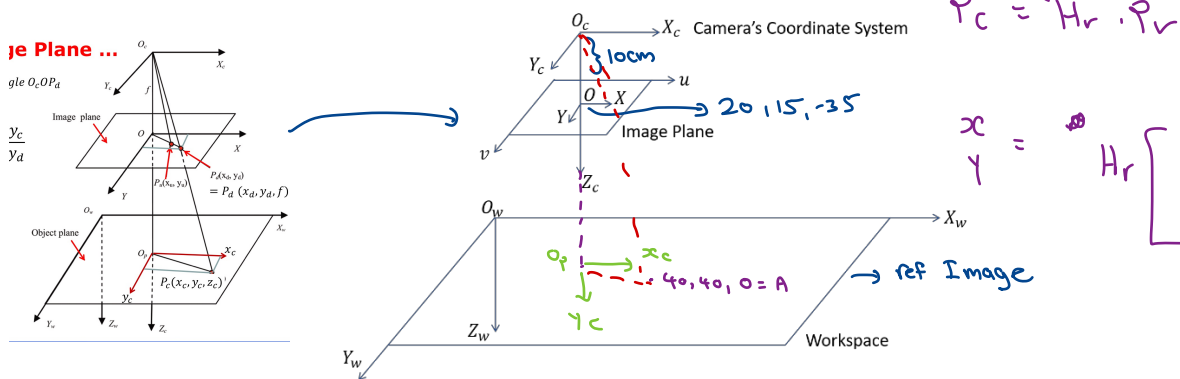
[. . .]

3) RCE architecture



Q3: → monocular vision

Refer to the figure below which shows a camera looking down at a workspace, which is parallel to the camera's image space. The distance between O and O_c is 10.0 cm. The coordinates of O are (20.0, 15.0, -35.0) (cm) with respect to the coordinate system $(OXYZ)_w$. If a point A is placed at (40.0, 40.0, 0.0) (cm) in $(OXYZ)_w$, what are the coordinates of point A 's image?



Your answer:

(4.44, 5.55)

Your Solution: (write the details of your solution here)

So 2 steps are involved! (when finding A's Image Coordinates)

1) Change coordinate from Scene to Camera

$$\begin{bmatrix} c_x \\ c_y \\ c_z \\ 1 \end{bmatrix} = {}^cH_r \begin{bmatrix} r_x \\ r_y \\ r_z \\ 1 \end{bmatrix} \rightarrow \text{maps ref frame to camera frame}$$

2) Change coordinate from Camera to Image (analogue)

$$\begin{pmatrix} s \cdot x \\ s \cdot y \\ 1 \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} c_x \\ c_y \\ c_z \\ 1 \end{pmatrix}$$

→ next page

3) we can combine the above equations

$$\begin{pmatrix} S \cdot x \\ S \cdot y \\ S \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

→ translation matrix

$$= \begin{bmatrix} fr_{11} & fr_{12} & fr_{13} & ft_x \\ fr_{21} & fr_{22} & fr_{23} & ft_y \\ fr_{31} & fr_{32} & fr_{33} & ft_z \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

→ rotation matrix

$$H_r = \begin{bmatrix} \cos(\theta^1)\cos(\theta^2) & -\cos(\theta^2)\sin(\theta^2) + \sin(\theta^2)\sin(\theta^1)\cos(\theta^3) & \sin(\theta^2)\sin(\theta^2) + \cos(\theta^2)\sin(\theta^1)\cos(\theta^3) & t_x \\ \cos(\theta^1)\sin(\theta^2) & \cos(\theta^2)\cos(\theta^2) + \sin(\theta^2)\sin(\theta^1)\cos(\theta^3) & -\sin(\theta^2)\sin(\theta^2) + \cos(\theta^2)\sin(\theta^1)\cos(\theta^3) & t_y \\ -\sin(\theta^1) & \sin(\theta^2)\cos(\theta^1) & \cos(\theta^2)\cos(\theta^1) & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

→ since directly parallel on top, so, $\theta^2 = -180^\circ$, $\theta^3, \theta^1 = 0$ } rotation of camera.

$$t_x = 20$$

$$t_y = 15$$

$$t_z = -35 - 10 \text{ (Since camera is -10 on top of image)}$$

$$\rightarrow \begin{bmatrix} -1 & 0 & 0 & 20 \\ -1 & 0 & 0 & 15 \\ 0 & 0 & 1 & -45 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} S \cdot x \\ S \cdot y \\ S \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{bmatrix} -1 & 0 & 0 & 20 \\ -1 & 0 & 0 & 15 \\ 0 & 0 & 1 & -45 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 40 \\ 40 \\ 0 \\ 1 \end{pmatrix}$$

$$= \begin{bmatrix} -10 & 0 & 0 & 200 \\ -10 & 0 & 0 & 150 \\ 0 & 0 & 1 & -45 \end{bmatrix} \begin{bmatrix} 40 \\ 40 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -200 \\ -250 \\ -45 \end{bmatrix}$$

$$S = -45$$

$$x = 4.44$$

$$y = 5.55$$

Index coordinates tells the location of that specific pixel with respect to the spatial coordinate.

Q4: Analogue to digital Image

Refer to Question 3. The index coordinates of O (i.e. the origin of the image plane's coordinate system OXY) are (511, 511). The image resolution (i.e. the image size in the index coordinate system uv) is 1024x1024. It is observed that when point A makes a displacement of (x=3.0, y=5.0) (cm), the index coordinates of its image make a displacement of (u=30, v=40) (pixels), what is the size of a pixel?

Your answer:

(0.00623, 0.0106)

Your Solution: (write the details of your solution here)

$$\begin{aligned} x, y &\rightarrow 3, 5 \\ u_0, v_0 &\rightarrow (511, 511) \\ (\Delta u, \Delta v) &\rightarrow ? \\ (u, v) &\rightarrow (30, 40) \end{aligned}$$

$$\begin{pmatrix} 30 \\ 40 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\Delta u} & 0 & 511 \\ 0 & \frac{1}{\Delta v} & 511 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$$

$$30 = \frac{3}{\Delta u} + 0 + 511 - (1) \quad (1)$$

$$40 = 0 + \frac{5}{\Delta v} + 511 - (2)$$

$$\Delta u = \frac{3}{511 - 30}$$

$$= 0.00623 \text{ cm/pixel}$$

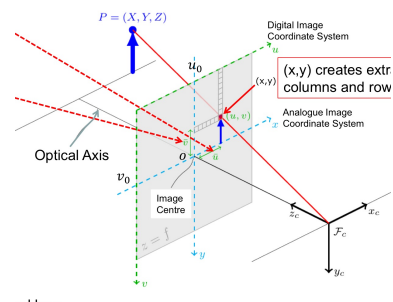
$$\Delta v = 0.0106 \text{ cm/pixel}$$

$$\begin{aligned} \text{Size} &\rightarrow (\Delta u, \Delta v) \\ &= 0.00623, 0.0106 \end{aligned}$$

Pixel loc Pixel Size

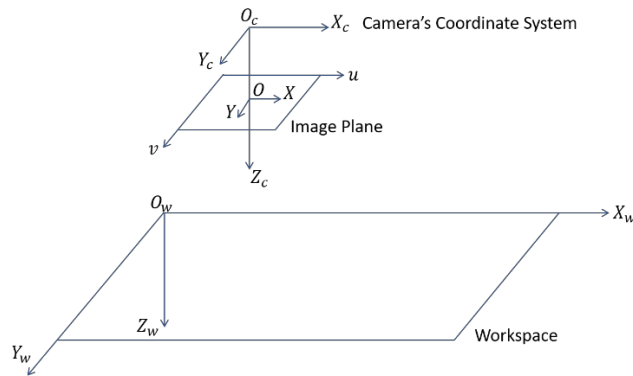
$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\Delta u} & 0 & u_0 \\ 0 & \frac{1}{\Delta v} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

origin (511, 511)
Spatial Coordinate



Q5:

Refer to the figure below. Now, the distance between O and O_c is 5.0 cm. The coordinates of O are (10.0, 20.0, -50.0) (cm) with respect to the coordinate system $(OXYZ)_w$. The image resolution is 2048x2048 while the index coordinates of O are (1023, 1023). Assume that the pixel size is 0.005x0.005 (cm). If an image processing algorithm detects a feature point at location (u=800, v=900), what are the coordinates of its object point in the plane $(OXY)_w$?



Your answer:

-1.115, -0.615

Your Solution: (write the details of your solution here)

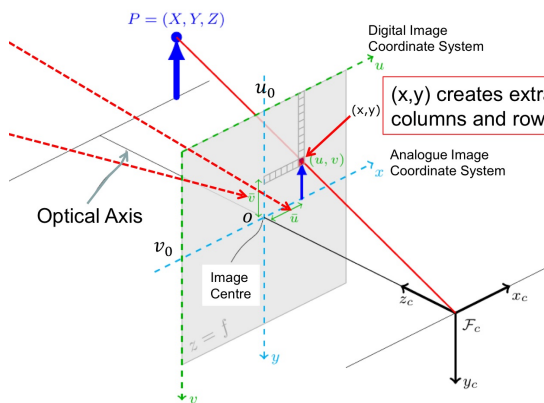
$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & u_0 \\ \Delta u & 0 & v_0 \\ 0 & \Delta v & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$x = \Delta u (v - v_0)$$

$$y = \Delta v (u - u_0)$$

$$x = 0.005 (800 - 1023) = -1.115$$

$$y = 0.005 (900 - 1023) = -0.615$$



$$\frac{20-25}{10-15}$$

$$20:10$$

-1

10

Q6:

Refer to Question 5. If a mobile robot moves from (10.0, 20.0, 0.0) (cm) to (15.0, 25.0, 0.0) (cm) by following a straight line, what will be the equation of the straight line inside the index coordinate system uv of the image plane?

Your answer:

$$V = U + 2000$$

Your Solution: (write the details of your solution here)

1) need to find uv coordinates first

initial point.

$$\begin{aligned} U &= \frac{x}{\Delta u} + U_0 \\ &= \frac{10}{0.005} + 1023 \\ &= 3023 \end{aligned}$$

$$\begin{aligned} V &= \frac{y}{\Delta v} + V_0 \\ &= \frac{20}{0.005} + 1023 \\ &= 5023 \end{aligned}$$

final pt

$$\begin{aligned} U &= \frac{15}{0.005} + 1023 \\ &= 4023 \end{aligned}$$

$$\begin{aligned} V &= \frac{25}{0.005} + 1023 \\ &= 6023 \end{aligned}$$

2) now find equation

$$V = aU + b$$

$$\begin{aligned} a &= \frac{V_2 - V_1}{U_2 - U_1} \\ &= \frac{5023 - 6023}{3023 - 4023} \\ &= 1 \end{aligned}$$

$$\begin{aligned} b &= V - aU \\ &= 6023 - 4023 \\ &= 2000 \quad \checkmark \end{aligned}$$

Q7:

Refer to Question 5. An image processing algorithm has detected a straight line in the index coordinate system uv of the image plane. The equation of the straight line is: $u = 2*v + 100$. What is the equation of the corresponding straight line in the workspace's coordinate system $(OXY)_w$?

Your answer:

$$y = 0.5x - 2.8$$

Your Solution: (write the details of your solution here)

$$u = \frac{x}{200} + u_0 \quad - (1)$$

$$v = \frac{y}{200} + v_0 \quad - (2)$$

$$200x + 1023 = 2(200y + 1023) + 100$$

$$200x + 1023 = 400y + 2043 + 100$$

$$400y = 200x + 1023 - 2043 - 100$$

$$400y = 200x - 1120$$

$$y = 0.5x - 2.8$$

Q8:

Refer to Question 5. An image processing algorithm has detected a circle in the index coordinate system uv of the image plane. The equation of the circle is: $(u - 1023)^2 + (v - 1023)^2 = 500^2$. What is the equation of the corresponding circle in the workspace's coordinate system $(OXY)_w$?

Your answer:

$$(200x)^2 + (200y)^2 = 500^2$$

Your Solution: (write the details of your solution here)

$$u = \frac{x}{\Delta u} + u_0 \quad - (1)$$

$$v = \frac{y}{\Delta v} + v_0 \quad - (2)$$

$$(u - 1023)^2 + (v - 1023)^2 = 500^2 \quad - (3)$$

Sub 1 & 2 into 3

$$\left(\frac{x}{0.005} + 1023 - 1023 \right)^2 + \left(\frac{y}{0.005} + 1023 - 1023 \right)^2 = 500^2$$

$$(200x)^2 + (200y)^2 = 500^2$$

Q9: → help

A 2D vision matrix is given as follows:

0.164499	0.022269	-27.176669
0.000767	-0.209450	74.306071
0.000018	0.001470	1.0

(m/pixel)

$$u=150, v=250$$

If an image processing algorithm has detected a feature point at the index coordinates ($u=150, v=250$), what are the coordinates of the corresponding object point in the workspace's coordinate system (OXY)_w?

Your answer:

716.3, -309.2

Your Solution: (write the details of your solution here)

$$u=150, v=250$$

$$S \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = M \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix}$$

$$S u = 0.164499 x_w + 0.022269 y_w - 27.176669 \quad (1)$$

$$S v = 0.000767 x_w - 0.209450 y_w + 74.306071 \quad (2)$$

$$S = 0.000018 x_w + 0.001470 y_w + 1.0 \quad (3)$$

Sub $u=150, v=250, (3)$ into (1) & (2)

for (1)

$$0.161799 x_w - 0.198231 y_w = 177.1766$$

for (2)

$$-0.003733 x_w - 0.576957 y_w = 175.693929$$

use simult eq.

$$x_w = 716.3$$

$$y_w = -309.2 //$$

Q10: $\rightarrow$ uses 2 Camera to map the 3D world

A binocular vision system has measured a surface in a global reference coordinate system. The measured surface can be described by the following equation in the reference coordinate system:

$$x + 2y + 3z + 4 = 0$$

If the binocular vision system outputs the coordinates of point A, which are (50.0, 10.0, -30.0) (cm), what is the distance from point A to the above planar surface?

Your answer:

4.276 cm

Your Solution: (write the details of your solution here)

$$\text{Distance} = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

$$= \frac{1(50) + 2(10) + 3(-30) + 4}{\sqrt{1^2 + 2^2 + 3^2}}$$

$$= \frac{|50 + 20 - 90 + 4|}{\sqrt{14}}$$

$$\approx 4.276 \text{ cm},$$